

A BAYESIAN APPROACH TO OUTLIER DETECTION IN REGRESSION ANALYSIS USING GIBBS SAMPLING

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Key words: Adaptive Monte Carlo integration technique; Bayesian approach; Gesell data; Gibbs sampling; outlier detection; posterior marginals; regression analysis.

Summary: The Bayesian analysis of outliers in a linear model is considered. Verdinelli and Wasserman (1991) showed that an adaptive Monte Carlo integration technique known as the Gibbs sampler is a conceptually and computationally simple method for calculating posterior marginals in outlier problems. The approach of Verdinelli and Wasserman is extended to the contaminated normal, location-scale model in the case of simple regression.

1. INTRODUCTION

The problem of outliers is studied and thoroughly reviewed by Barnett and Lewis (1984), Hawkins (1980), and Pettit and Smith (1985). The usual Bayesian approach to outlier detection uses the definition given by Freeman (1980). Freeman defines an outlier to be any observation that has not been generated by the mechanism that generated the majority of observations in the data set. Freeman's definition therefore requires that a model for the generation of outliers be specified.

The simplest example of this type of model is a finite mixture of normal distributions. Box and Tiao (1968), Guttman, Dutter and Freeman (1978), Pettit and Smith (1984, 1985) and Titterton, Smith and Makov (1985) consider such models. Calculating posterior marginals for an outlier model

typically involves difficult computations. All these authors comment on the computational problems in dealing with these models.

Verdinelli and Wasserman (1991) showed that an adaptive Monte Carlo integration technique known as the Gibbs sampler (Geman and Geman (1984), Casella and George (1990), Gelfand and Smith (1990, 1991) and Gelfand *et al* (1992)) is a conceptually and computationally simple method for calculating posterior marginals. By comparison with other techniques for finding posterior marginals (Achcar and Smith (1990) for example considered parameter transformations for successful implementation of Laplace's method) the Gibbs sampling approach is notable for its ease of implementation. The principal requirement of the sampler is that all conditional distributions must be available in the sense that random variates can be generated from them. For further details about the Gibbs sampler see for example Casella and George (1990), Gelfand and Smith (1990, 1991) and Gelfand *et al* (1992).

As mentioned by Verdinelli and Wasserman "The reason that the Gibbs sampler brings such simplicity to outlier problems is that the Gibbs sampler operates by iterating two different stages of calculations. In the first stage, we assume we know which observations are outliers. This assumption allows us to correct the outliers and we then sample from the posterior using the corrected data. In the second stage we assume that we know the true parameter values, and for each observation we compute the posterior probability that the observation is an outlier. This computation is simple since the parameters are assumed known. Alternating the two stages will eventually allow us to draw a sample from the posterior distribution. The important point is that the Gibbs sampler divides a difficult problem into a set of simple problems." In this note the approach of Verdinelli and Wasserman will be extended to regression analysis. In section 2 the contaminated, normal, location-scale model in the case of simple regression will be considered and in section 3 an example is given. In section 4 a sensitivity analysis is performed on some of the prior specifications of the model.

2. THE NORMAL LOCATION-SCALE PROBLEM IN THE CASE OF SIMPLE REGRESSION

We begin by considering the usual normal location-scale problem.

Let $\underline{y}' = (y_1, \dots, y_n)$ be an i.i.d. sample with density function

$$(2.1) \quad f(y_i | \beta_1, \beta_2, \sigma^2, \epsilon, \xi_i) = (1 - \epsilon) \phi(y_i | \beta_1 + \beta_2 X_i, \sigma^2) \\ + \epsilon \phi(y_i | \beta_1 + \beta_2 X_i + \xi_i, \sigma^2).$$

Let $\underline{\delta}' = (\delta_1 \dots \delta_n)$ be independent Bernoulli trials with success probability ϵ , and $\underline{\xi}' = (\xi_1 \dots \xi_n)$, then

$$(2.2) \quad y_i | \beta_1, \beta_2, \sigma^2, \underline{\xi}, \underline{\delta} \sim N(\beta_1 + \beta_2 X_i + \delta_i \xi_i; \sigma^2) \quad i = 1, \dots, n.$$

Note that each y_i is conditionally independent of ϵ . We will consider the case where ϵ is known.

In the Bayesian treatment of model (2.2) prior subjective uncertainty about β_1 and β_2 is often described via the "flat" prior density:

$$(2.3) \quad p(\beta_1, \beta_2) \propto \text{constant}.$$

The non-informative prior for the variance component is

$$(2.4) \quad p(\sigma^2) \propto \sigma^{-2} \quad 0 < \sigma^2 < \infty$$

and the proper priors for ξ_i and δ_i are

$$(2.5) \quad p(\xi_i | t) = \left[\frac{1}{2\pi t^2} \right]^{\frac{1}{2}} e^{-\xi_i^2/t^2} \quad -\infty < \xi_i < \infty$$

and

$$(2.6) \quad p(\delta_i | \epsilon) = \epsilon^{\delta_i} (1 - \epsilon)^{1 - \delta_i} \quad \delta_i = 0, 1$$

respectively for $i = 1, \dots, n$.

From (2.2) to (2.6) it follows that the joint density function can be written as

$$\begin{aligned} & f(y, \beta_1, \beta_2, \sigma^2, \xi, \underline{\delta}) \\ &= f(y | \beta_1, \beta_2, \sigma^2, \xi, \underline{\delta}) p(\beta_1, \beta_2) p(\sigma^2) p(\xi | t) p(\underline{\delta} | \epsilon) \\ &= \left[\frac{1}{2\pi\sigma^2} \right]^{\frac{1}{2}n} \exp \left\{ -\frac{1}{2} \sum_{i=1}^n (y_i - \delta_i \xi_i - \beta_1 - \beta_2 X_i)^2 / \sigma^2 \right\} \times \\ & \quad \left[\frac{1}{2\pi\sigma^2} \right]^{\frac{1}{2}n} \exp \left\{ -\frac{1}{2} \sum_{i=1}^n \xi_i^2 / t^2 \right\} \sigma^{-2} \prod_{i=1}^n \epsilon^{\delta_i} (1 - \epsilon)^{1 - \delta_i}. \end{aligned}$$

To implement the Gibbs-sampler we need the full conditionals

$$f_1(\beta | y, \sigma^2, \underline{\delta}, \xi, \epsilon), \quad f_2(\sigma^2 | y, \underline{\delta}, \xi, \epsilon), \quad f_3(\underline{\delta} | y, \beta, \sigma^2, \xi, \epsilon) \quad f_4(\xi | y, \beta, \sigma^2, \underline{\delta}, \epsilon),$$

where it is understood that f_3 is a probability mass function and $\underline{\beta}' = (\beta_1 \beta_2)$.

In order to obtain the density $f_1(\underline{\beta} | y, \sigma^2, \underline{\delta}, \xi, \epsilon)$ we consider $y^{*'} = (y_1^* \dots y_n^*)$ where $y_i^* = y_i - \delta_i \xi_i$.

Thus y^* is identical to the original data except that the outliers have been corrected by subtracting the location shift ξ_i . This correction is possible since both ξ and $\underline{\delta}$ are known at this step.

Now the y_i^* 's are independent samples from a $N(\beta_1 + \beta_2 X_i, \sigma^2)$ distribution with σ^2 known.

It therefore follows that

$$(2.7) \quad f_1(\underline{\beta} | \underline{y}, \sigma^2, \underline{\delta}, \underline{\xi}, \epsilon) = \left[\frac{1}{\sigma^2 2\pi} \right]^{\frac{1}{2}} |X'X|^{\frac{1}{2}} \exp \left\{ -\frac{1}{2} (\underline{\beta} - \hat{\underline{\beta}})' (X'X) (\underline{\beta} - \hat{\underline{\beta}}) / \sigma^2 \right\}$$

where

$$X' = \begin{bmatrix} 1 & 1 & \dots & 1 \\ X_1 & X_2 & \dots & X_n \end{bmatrix}, \quad \hat{\underline{\beta}} = (X'X)^{-1} X' \underline{y}^*,$$

$$f_2(\sigma^2 | \underline{y}, \underline{\beta}, \underline{\delta}, \underline{\xi}, \epsilon) = \frac{\left\{ \sum_{i=1}^n (y_i - \delta_i \xi_i - \beta_1 - \beta_2 X_i)^2 \right\}^{\frac{1}{2}n}}{\Gamma\left[\frac{n}{2}\right] 2^{n/2}} \times$$

$$(2.8) \quad (\sigma^{-2})^{\frac{1}{2}n+1} \exp \left\{ -\frac{1}{2} \sum_{i=1}^n (y_i - \delta_i \xi_i - \beta_1 - \beta_2 X_i)^2 / \sigma^2 \right\}$$

and

$$(2.9) \quad f_3(\underline{\delta} | \underline{y}, \underline{\beta}, \sigma^2, \underline{\xi}, \epsilon) = \prod_{i=1}^n p_i^{\delta_i} (1-p_i)^{1-\delta_i} \quad \delta_i = 0, 1$$

where

$$p_i = \frac{\epsilon e^{-\frac{1}{2}(y_i - \xi_i - \beta_1 - \beta_2 X_i)^2 / \sigma^2}}{\epsilon e^{-\frac{1}{2}(y_i - \xi_i - \beta_1 - \beta_2 X_i)^2 / \sigma^2} + (1-\epsilon) e^{-\frac{1}{2}(y_i - \beta_1 - \beta_2 X_i)^2 / \sigma^2}}$$

From f_3 it therefore follows that conditional on the data and the other parameters, each δ_i is independently Bernoulli distributed with success probability p_i .

Finally consider f_4 . For $\delta_i = 1$,

$$(2.10a) \quad \xi_i | y, \underline{\beta}, \sigma^2, \delta_i = 1 \sim N(c, d)$$

where

$$c = \frac{(y_i - \beta_1 - \beta_2 X_i) / \sigma^2}{1/t^2 + 1/\sigma^2} \text{ and } d = (1/t^2 + 1/\sigma^2)^{-1}$$

and for $\delta_i = 0$ it follows that

$$(2.10b) \quad \xi_i | y, \underline{\beta}, \sigma^2, \delta_i = 0 \sim N(0, t^2).$$

In other words if $\delta_i = 0$, then we have no information on ξ_i (that is the likelihood for ξ_i is flat) so that the conditional distribution $\xi_i | y, \underline{\beta}, \sigma^2, \delta_i = 0$ is simply its prior distribution, namely, $N(0, t^2)$. If one is interested in an improper prior for ξ_i , this can be approximated in the algorithm by using a large t -value.

Using (2.1) it also follows that

$$(2.11) \quad (\beta_1 + \beta_2 X_i) | y, \sigma^2, \underline{\delta}, \underline{\xi}, \epsilon \sim N(\hat{\beta}_1 + \hat{\beta}_2 X_i, h_i \sigma^2)$$

and

$$(2.12) \quad u_i | y, \sigma^2, \underline{\delta}, \underline{\xi}, \epsilon \sim N(\hat{u}_i, h_i \sigma^2)$$

where $u_i = y - \beta_1 - \beta_2 X_i$, $\hat{u}_i = y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i$

and $h_i = (1 \ X_i)(X'X)^{-1}(1 \ X_i)'$, the i -th diagonal element of the hat

matrix $H = X(X'X)^{-1}X'$. Equation (2.11) is the conditional posterior distribution of the response variable for X_i , a specific value of the independent variable and (2.12) is the conditional posterior distribution of the i -th residual.

The Gibbs sampler can now easily be implemented. Standard routines can be used to generate random numbers from the required distribution. The iterative process starts by using arbitrary starting values $\sigma^{2(0)}$, $\underline{\delta}^{(0)}$, $\xi^{(0)}$ in (2.7) to draw $\underline{\beta}^{(1)}$. To complete the first iteration, these values are put in (2.8) through (2.10) to draw $\sigma^{2(1)}$, $\underline{\delta}^{(1)}$ and $\xi^{(1)}$. After k iterations in which the conditional distributions are updated at each iteration the Gibbs sampler has generated the values $\underline{\beta}^{(k)}$, $\sigma^{2(k)}$, $\underline{\delta}^{(k)}$ and $\xi^{(k)}$. The process is then repeated m times.

The convergence of the algorithm is discussed in Gelfand and Smith (1990). Various techniques have been developed to check the convergence to the marginal distributions. See, for example, Zeger and Karim (1989) and Carlin, Gelfand and Smith (1989). This convergence issue remains an area of active research. Verdinelli and Wasserman (1991) suggested that m should be fairly large but that k need not be large. Unless otherwise mentioned the examples in this paper were performed with $k = 30$ and $m = 1\,000$. In the case of the sensitivity analysis (section 4), $k = 20, 40$ and 500 were also used. The posterior marginal for $\underline{\beta}$ may be estimated as

$$(2.13) \quad f(\underline{\beta} | \underline{y}) = \frac{1}{m} \sum_{j=1}^m \phi(\underline{\beta} | (\hat{\underline{\beta}})_j^{(k)}, (X'X)^{-1}(\sigma^2)_j^{(k)}).$$

Similarly the posterior probability that y_i is an outlier is estimated by

$$(3.14) \quad \frac{1}{m} \sum_{j=1}^m \frac{e^{-\frac{1}{2}(y_i - (\xi_i)_j^{(k)} - (\beta_1)_j^{(k)} - (\beta_2)_j^{(2)})^2 / (\sigma_j^2)^{(k)}}}{e^{-\frac{1}{2}(y_i - (\xi_i)_j^{(k)} - (\beta_1)_j^{(k)} - (\beta_2)_j^{(k)})^2 / (\sigma_j^2)^{(k)}} + (1-\epsilon)e^{-\frac{1}{2}(y_i - (\beta_1)_j^{(k)} - (\beta_2)_j^{(k)})^2 / (\sigma_j^2)^{(k)}}}$$

The posterior for σ^2 is less interesting and we do not bother to estimate it.

3. EXAMPLE

The Gibbs sampler will now be illustrated by using the Gesell adaptive score data (Mickey, Dunn and Clark (1967)).

The first three columns of table 3.1 show a data set on cyanatic heart disease in children. The response variable y is the child's Gesell adaptive score, and the only predictor variable X is the child's age in months at first word. These data were first presented by Mickey *et al.* (1967) and have been used by several authors to evaluate the performance of various regression diagnostics.

TABLE 3.1 Gesell adaptive score data: X_i is age, in months, at first word and y_i is adaptive score.

i	X_i	y_i	i	X_i	y_i	i	X_i	y_i
1	15	95	8	11	100	15	11	102
2	26	71	9	8	104	16	10	100
3	10	83	10	20	94	17	12	105
4	9	91	11	7	113	18	42	57
5	15	102	12	9	96	19	17	121
6	20	87	13	10	83	20	11	86
7	18	93	14	11	84	21	10	100

We used $\epsilon = 0,05$ and $t = 1\,000$. This value of ϵ is typically regarded as being reasonable for these data (see for example Pettit (1992)). The large value of t is used to reproduce the usual flat prior used in this problem.

The estimated posterior probabilities that each observation is an outlier is given in table 3.2.

TABLE 3.2: Posterior probability that an observation is an outlier for the Gesell adaptive score data.

i	Probability	i	Probability	i	Probability
1	0,008	8	0,007	15	0,005
2	0,003	9	0,010	16	0,004
3	0,050	10	0,001	17	0,015
4	0,004	11	0,002	18	0,004
5	0,004	12	0,009	19	0,817
6	0,006	13	0,027	20	0,008
7	0,001	14	0,012	21	0,002

From table 3.2 it can be seen that only observation 19 shows up as an outlier, which is in agreement with the conclusion of previous studies (Chaloner and Brant (1988) and Pettit (1992)). From equation (2.12) it is clear that the Gibbs sampler also provides the unconditional posterior distributions of the residuals $u_i = y_i - \beta_1 - \beta_2 X_i$. Interval estimates of u_i are easily constructed. These intervals can be added to a residual plot. The plots can be thought of as describing the marginal posterior distributions of the u_i . In figure 3.1 the residual plot of $E(u_i|y)$ against the case number, i , with the 95% highest posterior density intervals are given.

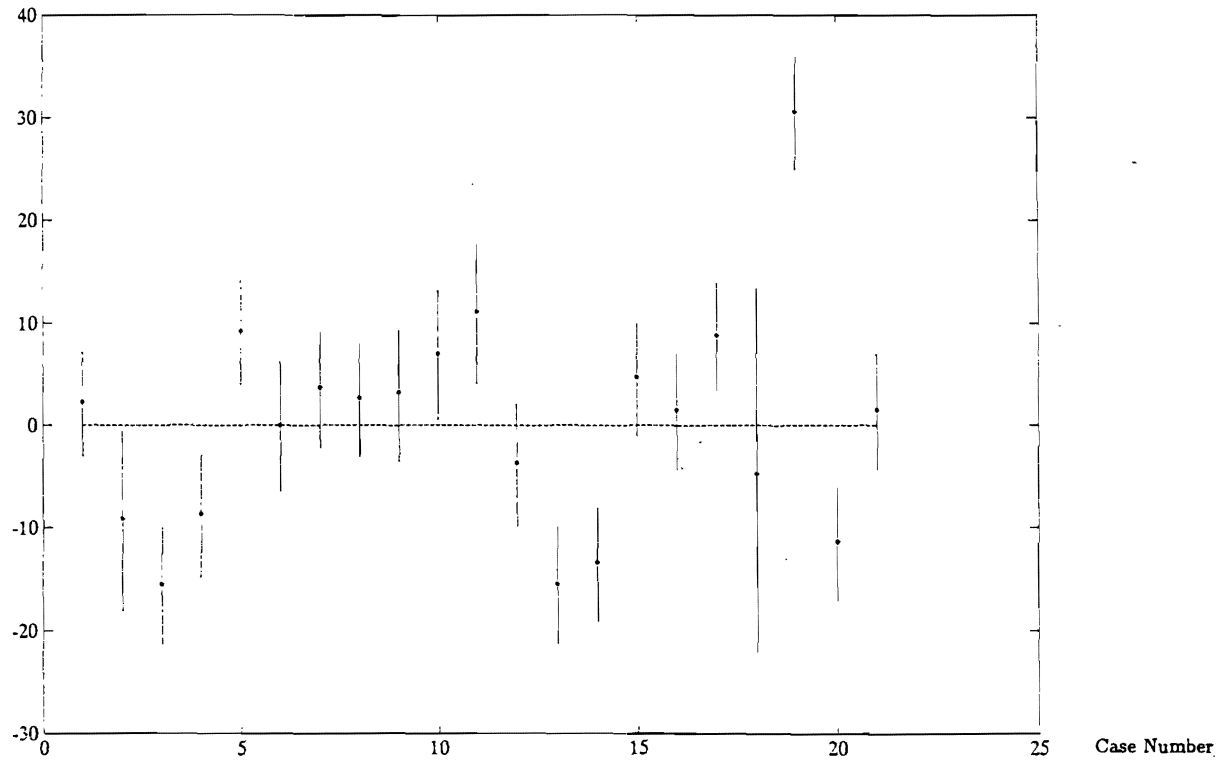


FIGURE 3.1: Plot of the expected posterior residual against case number i , with the 95% highest posterior density interval

The plot provides a simple summary of the information in the data about the u_i 's. Observation 19 is noticeable as a point with large $E(u_i|\mathbf{y})$ and small posterior variance, while observation 18 has a residual with large posterior variance. Observation 18 has the largest leverage, h_{18} , and although there is uncertainty about u_{18} , the observation need not be regarded an outlier. The observation is, however, very influential. All other observations (except observation 19) have posterior probability of being an outlier equal to or less than 0.05.

Using equation (2.11) and the Gibbs sampler, the unconditional posterior distribution of the i -th response variable $(\beta_1 + \beta_2 X_i)$ can also be obtained. In figure 3.2 the distributions for observations 2, 18 and 19 are displayed.

Also from this figure it is evident that, although observation 19 is an outlier, it has a small posterior variance, while observation 18 has a large variance.

In table 3.3 the unconditional posterior means $E(\beta_1 + \beta_2 X_i|\mathbf{y})$ are compared with $\hat{y}_i = \hat{\beta}_1 + \hat{\beta}_2 X_i$ and the true values y_i .

From the table it is clear that there is not much difference between $\hat{y}_i$ and $E(\beta_1 + \beta_2 X_i|\mathbf{y})$. Both methods fail to estimate the observed value in the case of observation 19. The reason for this is that observation 19 can be considered to be an outlier, i.e. coming from a normal population with larger mean. Using the Gibbs sampler we can also calculate $E(\beta_1 + \beta_2 X_{19} + \xi_{19}|\mathbf{y})$ which is equal to 121.93 and near the observed value of 121.

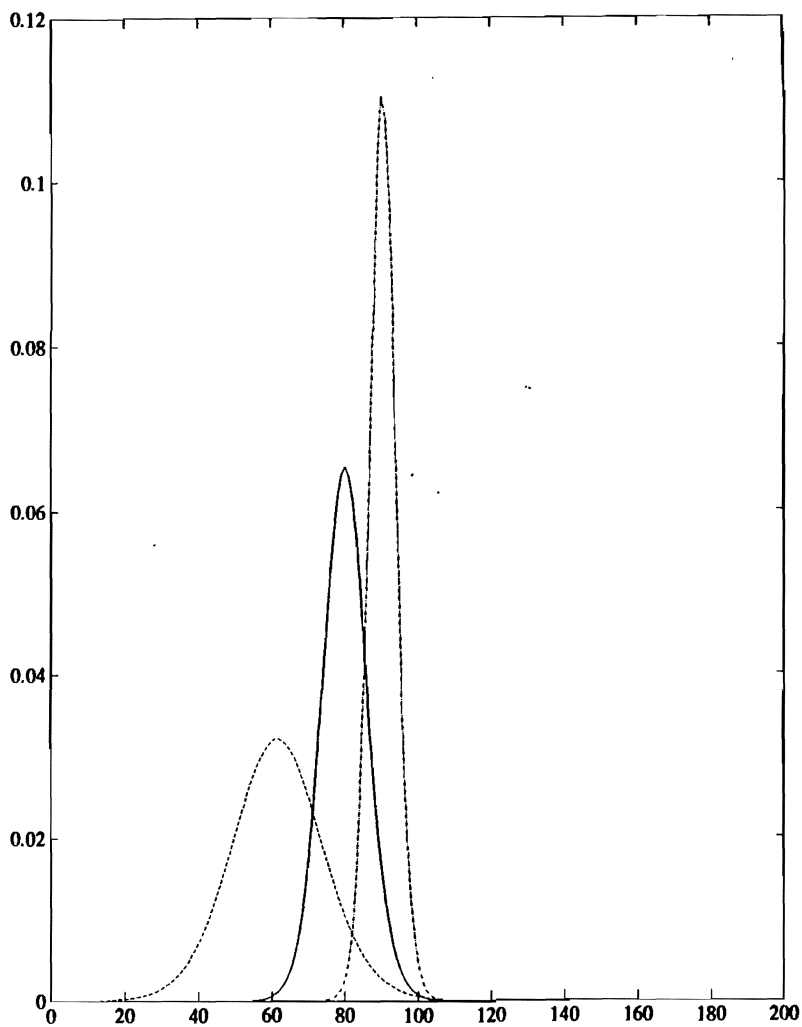


FIGURE 3.2: Unconditional posterior distributions of $\beta_1 + \beta_2 X_i$

— Observation 2 --- Observation 18 -.-.- Observation 19

TABLE 3.3: Unconditional posterior means $E(\beta_1 + \beta_2 X_i | y)$ and least squares estimates $\hat{y}_i = \hat{\beta}_1 + \hat{\beta}_2 X_i$ for Gesell adaptive score data

Observation	$E(\beta_1 + \beta_2 X_i y)$	$\hat{y}_i$	y_i	Observation	$E(\beta_1 + \beta_2 X_i y)$	$\hat{y}_i$	y_i
1	93,12	92,97	95	12	99,71	99,73	96
2	81,05	80,57	71	13	98,61	98,60	83
3	98,61	98,60	83	14	97,51	97,48	84
4	99,71	99,73	91	15	97,51	97,48	102
5	93,12	92,97	102	16	98,61	98,60	100
6	87,64	87,33	87	17	96,41	96,35	105
7	89,83	89,59	93	18	63,49	62,54	57
8	97,51	97,48	100	19	90,93	90,72	121
9	100,80	100,86	104	20	97,51	97,48	86
10	87,64	87,33	94	21	98,61	98,60	100
11	101,90	101,98	113				-

4. SENSITIVITY ANALYSIS

In this section a sensitivity analysis is performed on the weakly informative values of t and ϵ . Posterior densities are also compared for $k=20, 30$ and 40 iterations as well as for an alternative Markov Chain Monte Carlo (MCMC) procedure in which a "burn-in period" of $k=500$ is used. In this case only $(1\,000 + k) \times 4 = 6\,000$ generations and substitutions are required. For the usual procedure the number of generations are $30 \times 4 \times 1\,000 = 120\,000$. In this sense the Gibbs sampling procedure is computationally expensive.

In figures 4.1 and 4.2 the unconditional posterior densities of the response variable $(\beta_1 + \beta_2 X_1)$ are displayed for cases 2, 18 and 19 for different values of k . From the figures it is clear that the densities are very similar. Also the posterior probabilities that observation 19 is an outlier are for all practical purposes the same: 0,793; 0,806 (0,817); 0,812 and 0,723 for $k=20, 30, 40$ and 500 respectively. The number in brackets is the value of the posterior probability obtained from another run of $m=1\,000$ and $k=30$ generations. (See table 3.2.)

It therefore seems certain that convergence has been achieved in all four cases and that the MCMC approach is applicable here. Gelman and Rubin (1992), however, cautioned that a single series from the Gibbs sampler can provide a false sense of security.

A sensitivity analysis is also performed on the weakly informative values of t . In figures 4.3 and 4.4 the posterior densities of the response variable are again displayed but for different values of t (50, 100, 500 and 1 000). It can be seen that the densities are less peaked, i.e. flatter for larger values of t . This is obvious but what is perhaps less clear is why the posterior probability (that observation 19 is an outlier) is increasing with increasing values of t . For $t=50, 100, 500$ and $1\,000$, the posterior probabilities are 0,616; 0,696; 0,764 and 0,806 respectively. The reason for this is that for small t more observations are forced to become potential outliers with the result that the posterior probability of observation 19 decreases relative to

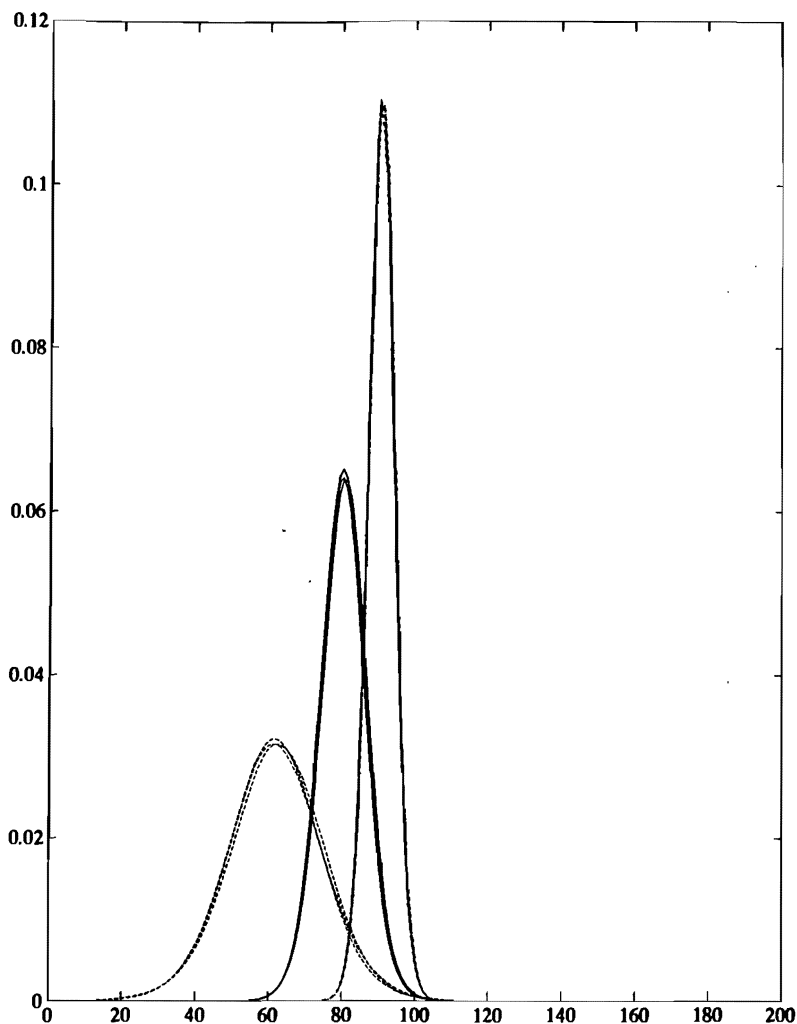


FIGURE 4.1: A comparison of the unconditional posterior distributions of $\beta_1 + \beta_2 X_i$ for various values of k
 $k = 20$ – Upper density; $k = 30$ – Lower density;
 $k = 40$ – The density in between

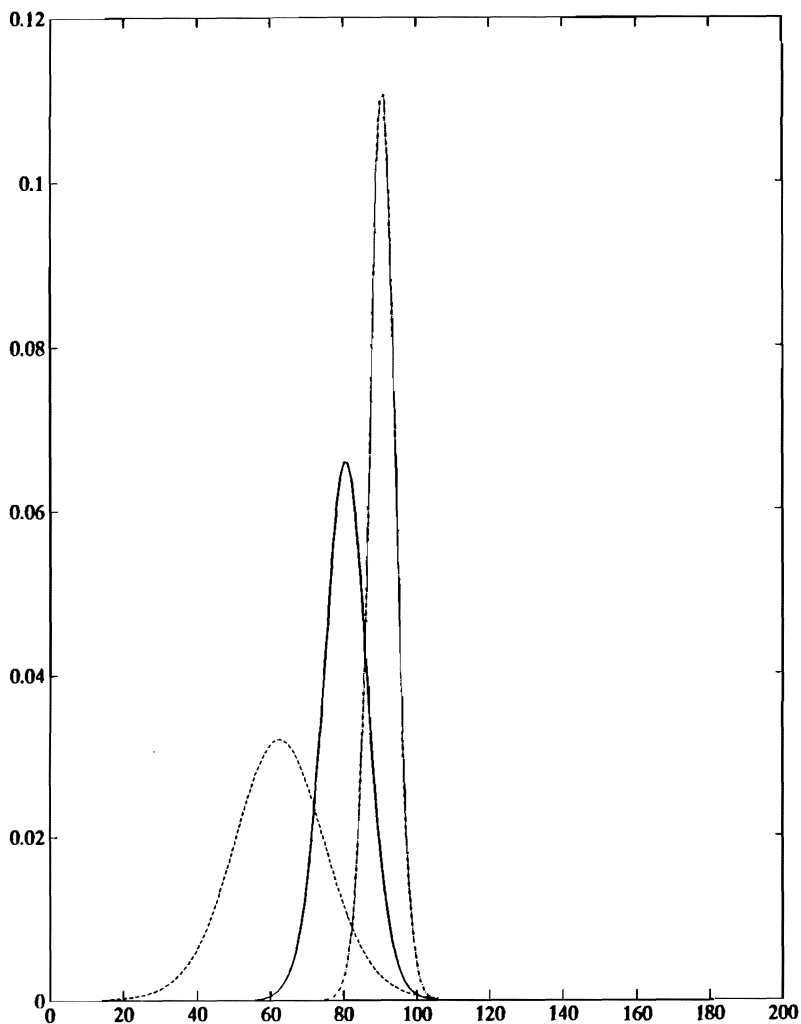


FIGURE 4.2: Unconditional posterior distribution of $\beta_1 + \beta_2 X_i$ for $k = 500$ in the case of the MCMC procedure

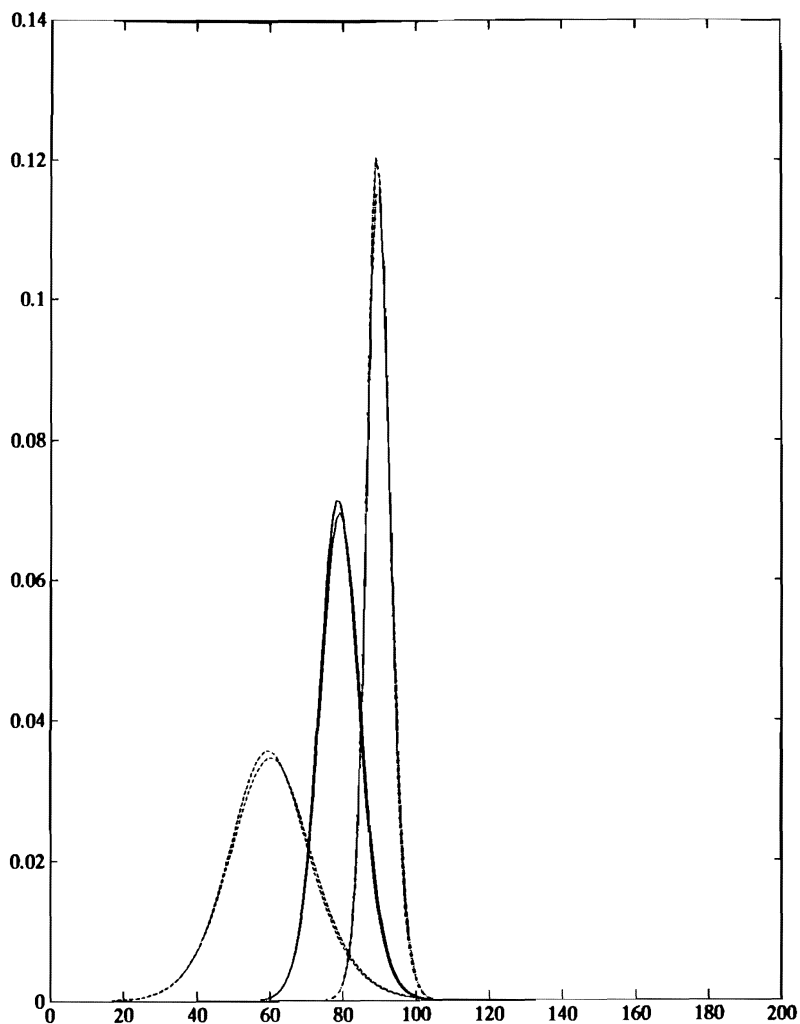


FIGURE 4.3: Unconditional posterior distributions of $\beta_1 + \beta_2 X_i$
for various values of t $t = 50$ – upper density;
 $t = 100$ – lower density

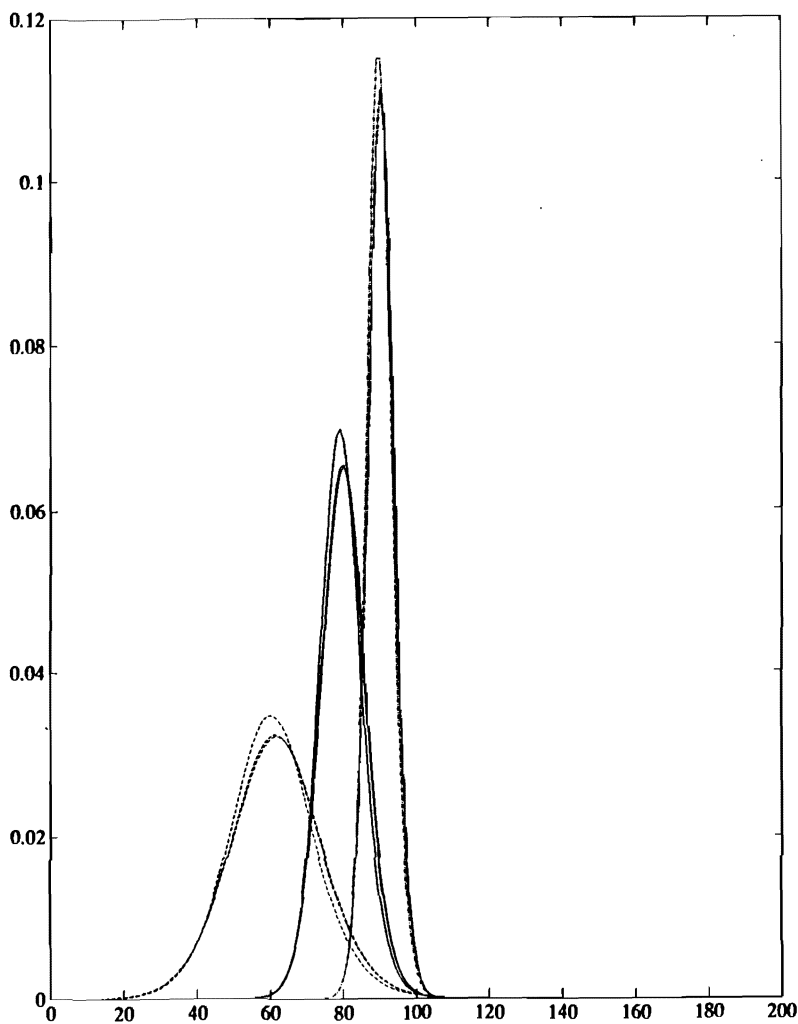


FIGURE 4.4: Unconditional posterior distributions of $\beta_1 + \beta_2 X_i$ for various values of t
 $t = 100$ – upper density; $t = 500$ – the density in between;
 $t = 1\,000$ – lower density

the other observations. t should therefore be large. A large t will ensure that the "real" outliers have large posterior probabilities and will also prevent misspecification of the priors of ξ_i ($i = 1, \dots, n$).

Let us now consider the case when ϵ is unknown. The densities f_1, f_2, f_3 and f_4 remain unchanged since they are conditional on ϵ ; but a fifth conditional posterior distribution is needed namely the distribution for ϵ conditional on the data and the rest of the parameters. As in Verdinelli and Wasserman it is assumed that ϵ has a Beta (ρ_1, ρ_2) prior distribution. A flat prior is not appropriate because the experiment would not have been conducted if there was an exceedingly large probability of having many outliers. As mentioned before, several authors have considered a value of 0,05 to be reasonable for ϵ . The mean of the prior for ϵ is therefore taken to be 0,05. It also seems reasonable to assume that an observation has less than half a chance of being an outlier with a high probability. If it is assumed that $P(\epsilon < 0,5) = 0,99$ then this implies that $\rho_1 = 0,1842$ and $\rho_2 = 3,5$. It is readily seen that the conditional distribution for ϵ depends only on δ . Therefore let r be the number of δ_i 's that are equal to one; then we simply have a binomial experiment with r successes and a Beta prior. Hence the required conditional posterior density for ϵ is Beta($\rho_1 + t, \rho_2 + n - r$) from which random numbers can easily be generated. The posterior marginal for ϵ is estimated by

$$(4.1) \quad \hat{f}(\epsilon | y) = \frac{1}{m} \sum_{j=1}^m B(\rho_1 + \tau(j), \rho_2 + n - \tau(j))$$

where $B(\rho_1, \rho_2)$ is the density for a Beta (ρ_1, ρ_2) distribution and $\tau(j)$ is the number of the $\{(\delta_1)_j^{(k)}, \dots, (\delta_n)_j^{(k)}\}$ that are equal to one.

For ϵ unknown the posterior probability that observation 19 is an outlier has dropped to 0,681. This slight decrease suggests that uncertainty about ϵ lowers the probability that observation 19 is an outlier.

Specifically, since small values of ϵ are possible the posterior probability that this observation is an outlier is lowered, though it still stands out as a likely outlier relative to the other observations.

The posterior distributions for the other parameters remain essentially the same as in the known ϵ case.

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